* Your Java program must be able to connect to the PostgreSQL server on cs1.calstatela.edu! You WON'T get credit if it connects to your local PostgreSQL server!

* Your submissions should include

1. A doc or pdf file: ER diagram, the database schema diagram, SQL statements used to create the tables, the source code of the stored procedures and triggers, and the percentage of each member's contribution to the project.
2. Java source code in a java file.
3. **A list of each member's contributions along with the corresponding percentage for each.**

* This is a group project. One submission per group!

Please design and implement the following university database application:

a. Professors have an SSN, a name, an age, gender, a rank, and a research specialty.

b. Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget.

c. Graduate students have an SSN, a name, an age, gender, and a degree program (e.g., M.S.or Ph.D.).

d. Each project is managed by exactly one professor (known as the project's principal investigator).

e. Each project is worked on by one or more professors (known as the project's co-investigators).

f. Professors can manage and/or work on multiple projects.

g. Each project is worked on by one or more graduate students (known as the project's research assistants).

h. Departments have a department number, a department name, and a main office.

i. Departments must have one professor (known as the chairman) who runs the department.

j. Professors work in exactly one department

k. Graduate students have one major department in which they are working on their degree.

l. Each graduate student is assigned a more senior graduate student, known as a student advisor, who provides guidance on course selection

1. (25pt) Design a conceptual schema and draw an ER diagram for your schema. Be sure to indicate all key and cardinality constraints and any assumptions you make. Identify any constraints you are unable to capture in the ER diagram and briefly explain why you could not express them.

2. (25pt) Translate the ER diagram into tables. Write SQL statements to create the tables and specify the constraints. Populate the tables with some sample data

3. (20pt) Triggers and Stored Procedures: Add the following triggers and stored procedures to your database:

a. female_faculty(): a function that returns the percentage of the female faculty in the university.

b. total_people(pno): a function that returns the total number of people (PI, CO-PI, students) who work on the project with pno.

c. faculty_restrict: a trigger to ensure that each project has at most 4 faculty members as CO-PI.

d. student_restrict: a trigger to ensure that a student works on no more than two projects

4. (30pt) Java and JDBC: Write a Java client to the database such that a person who does not know SQL could perform the following tasks:

- Add /remove a project
- Display the information of a student, the name of his/her student advisor, and the major department.
- Display the information of all projects that a faculty works on

Project Rubric					
Criteria	**Ratings**				**Points**
Part I	**Full Marks** Correctly identify all the entity types, relationship types, and constraints. **25 pts**	**Above half marks** Correctly identify 80% of the entity types, relationship types, and constraints. **20 pts**	**Half marks** Correctly identify 40% of the entity types, relationship types, and constraints. **12 pts**	**Below half marks** Failed to correctly identify most of the entity types, relationship types, and constraints. **5 pts**	/25 pts
Part II	**Full Marks** All entity types and relationship types are correctly mapped to tables. All constraints are reflected during the mapping. **25 pts**	**Above half marks** Most of the entity types and relationship types are correctly mapped to tables. Most of the constraints are reflected during the mapping. **20 pts**	**Half marks** Half of the entity types and relationship types are correctly mapped to tables. Half of the constraints are reflected during the mapping. **12 pts**	**Below half marks** Most of the entity types and relationship types are not correctly mapped to tables. Most of the constraints are not reflected during the mapping. **5 pts**	/25 pts
Part III	**Full Marks** The 2 funstions and the 2 triggers work correctly **20 pts**	**Above half marks** Only one trigger or one function doesn't work properly. **15 pts**	**Half marks** Half of the functions or triggers don't work properly. **10 pts**	**Below half marks** Most of the funstions or triggers don't work properly. **5 pts**	/20 pts
Part IV	**Full Marks** The application works. All the tasks are correctly implemented. **30 pts**	**Above half marks** The application works in most cases. Most of the tasks are correctly implemented. **20 pts**	**Half marks** The application doesn't work in some cases. Only half of the tasks are correctly implemented. **15 pts**	**Below half marks** The application doesn't work. **10 pts**	/30 pts